

Joint Same-Centre Fusco–Julin Stability and Borel Measurable Centre Selection (Plan 2, Deliverables B + C)

Abstract

This note delivers a single *joint* same-centre Fusco–Julin stability statement, by minimising the combined functional $\Phi_E(z) := \alpha_{\text{FJ}}(E, z)^2 + b_{\text{FJ}}(E, z)^2$ over a fixed compact admissible centre set $K \subset \mathbb{R}^n$ containing the unit ball $\overline{B}_2$. For sets $E \subset K$ of unit volume with small isoperimetric deficit $\mathcal{D}(E)$, we prove

$$\min_{z \in K} \Phi_E(z) \leq C_{\text{scFJ}}(n) \mathcal{D}(E),$$

unconditionally on $n \geq 2$. The proof combines: the Figalli–Maggi–Pratelli sharp quantitative isoperimetric inequality (α_{FJ} -control at the Fraenkel centre); the Fusco–Julin sharp normal-oscillation estimate (b_{FJ} -control at a separate FJ centre); a quantitative “centres are close” estimate $|z_F - z_{\text{FJ}}| \leq C \mathcal{D}(E)^{1/2}$ derived from the Fuglede nearly-spherical parametrisation; and a one-line change-of-centre inequality for the Fraenkel asymmetry. We then upgrade attainment and constancy of the bound to a Borel measurable selector $\rho \mapsto z_\rho$ on the good set of torsion levels.

The theorem is the exact import needed by the `WEIGHTEDMETRICTRACE` and `FJCENTERBRIDGE` blocks and replaces the conditional Theorem 1.2 of `WIP_FJCenterBridge.tex`.

1 Setup and notation

Throughout, $n \geq 2$ is fixed, $\omega_n = |B_1|$, and $P(B_1) = n\omega_n$. For a set of finite perimeter $E \subset \mathbb{R}^n$ with $0 < |E| < \infty$ and a point $z \in \mathbb{R}^n$, write

$$R_E := (|E|/\omega_n)^{1/n}, \quad B_E(z) := B_{R_E}(z), \quad P_E^* := P(B_E(z)) = n\omega_n R_E^{n-1},$$

and define the *pointed* (centred) asymmetry and normal oscillation

$$\alpha_{\text{FJ}}(E, z) := \frac{|E \Delta B_E(z)|}{|B_E(z)|}, \quad \beta_{\text{FJ}}(E, z)^2 := \frac{1}{2} \int_{\partial^* E} \left| \nu_E(x) - \frac{x-z}{|x-z|} \right|^2 d\mathcal{H}^{n-1}(x),$$

with the value of the radial vector field at $x = z$ chosen arbitrarily (the singleton $\{z\}$ is $\mathcal{H}^{n-1}$ -null for $n \geq 2$). The scale-invariant version is $b_{\text{FJ}}(E, z)^2 := \beta_{\text{FJ}}(E, z)^2 / P_E^*$. The isoperimetric deficit is $\mathcal{D}(E) := (P(E) - P_E^*) / P_E^*$.

When restricted to sets of unit volume, $R_E = 1$ and $B_E(z) = B_1(z)$. The Fraenkel asymmetry and the FJ oscillation, with their separately optimal centres, are

$$\alpha(E) := \inf_{z \in \mathbb{R}^n} \alpha_{\text{FJ}}(E, z), \quad \beta(E) := \inf_{z \in \mathbb{R}^n} \beta_{\text{FJ}}(E, z).$$

We will work with the *joint pointed functional*

$$\Phi_E(z) := \alpha_{\text{FJ}}(E, z)^2 + b_{\text{FJ}}(E, z)^2, \quad z \in \mathbb{R}^n.$$

2 Separately-minimised inputs

We use two well-established statements as black boxes.

Theorem 2.1 (Figalli–Maggi–Pratelli sharp isoperimetric stability; [3, 6]). *There exists $C_F = C_F(n) > 0$ such that every set $E \subset \mathbb{R}^n$ of finite perimeter with $|E| = \omega_n$ satisfies*

$$\alpha(E)^2 \leq C_F \mathcal{D}(E).$$

The infimum is attained: there exists $z_F(E) \in \mathbb{R}^n$ with $\alpha_{FJ}(E, z_F) = \alpha(E)$ (attainment follows from L^1 -continuity of translation and tightness for small deficit; cf. [7, Ch. 12–13]).

Theorem 2.2 (Fusco–Julin sharp normal oscillation; [5]). *There exists $C_{FJ} = C_{FJ}(n) > 0$ such that every set $E \subset \mathbb{R}^n$ of finite perimeter with $|E| = \omega_n$ satisfies*

$$\beta(E)^2 \leq C_{FJ} \mathcal{D}(E).$$

The infimum is attained: there exists $z_{FJ}(E) \in \mathbb{R}^n$ with $b_{FJ}(E, z_{FJ}) = \beta(E)$; attainment follows from continuity in z (Lemma 3.1) and a coercivity argument on a compact admissible set after a Fraenkel localisation (see Lemma 3.3).

We will only use Theorems 2.1 and 2.2 as *separately-minimised* bounds; we do not assume they hold at a common centre.

3 Joint same-centre theorem

3.1 Continuity of Φ_E in the centre

Lemma 3.1 (Continuity in the centre). *For every set E of finite perimeter with $0 < |E| < \infty$ and $P(E) < \infty$, the map $z \mapsto \Phi_E(z)$ is continuous on $\mathbb{R}^n$.*

Proof. We treat the two summands separately.

Continuity of $z \mapsto \alpha_{FJ}(E, z)$. The map $z \mapsto \mathbf{1}_{B_E(z)}$ is continuous from $\mathbb{R}^n$ to $L^1(\mathbb{R}^n)$: for any $z_k \rightarrow z$, $\|\mathbf{1}_{B_E(z_k)} - \mathbf{1}_{B_E(z)}\|_{L^1} = |B_E(z_k) \Delta B_E(z)| \rightarrow 0$ by translation continuity of the indicator in L^1 . Hence $z \mapsto |E \Delta B_E(z)| = \|\mathbf{1}_E - \mathbf{1}_{B_E(z)}\|_{L^1}$ is continuous, and therefore so is its square.

Continuity of $z \mapsto \beta_{FJ}(E, z)^2$. For $z_k \rightarrow z$ define

$$g_k(x) := \left| \nu_E(x) - \frac{x - z_k}{|x - z_k|} \right|^2, \quad g(x) := \left| \nu_E(x) - \frac{x - z}{|x - z|} \right|^2.$$

For $\mathcal{H}^{n-1}$ -a.e. $x \in \partial^* E$, $x \neq z_k$ for large k and $x \neq z$, so the radial unit vector is continuous at (x, z) ; hence $g_k(x) \rightarrow g(x)$ pointwise $\mathcal{H}^{n-1}$ -a.e. on $\partial^* E$. Since each g_k is uniformly bounded by 4 and $\mathcal{H}^{n-1}(\partial^* E) = P(E) < \infty$, dominated convergence gives $\beta_{FJ}(E, z_k)^2 \rightarrow \beta_{FJ}(E, z)^2$. Division by $2P_E^*$, a constant in z , preserves continuity.

The sum Φ_E is thus continuous in z . □

3.2 Change-of-centre estimate for the Fraenkel asymmetry

Lemma 3.2 (Change of Fraenkel centre). *For every set $E \subset \mathbb{R}^n$ of finite perimeter with $|E| = \omega_n$, and every $z, z' \in \mathbb{R}^n$,*

$$|\alpha_{FJ}(E, z) - \alpha_{FJ}(E, z')| \leq \frac{|B_1(z) \Delta B_1(z')|}{\omega_n} \leq \frac{n \omega_{n-1}}{\omega_n} |z - z'|.$$

Proof. By the triangle inequality for symmetric differences, $|E\Delta B_1(z)| \leq |E\Delta B_1(z')| + |B_1(z')\Delta B_1(z)|$, and the same with z, z' interchanged; dividing by ω_n gives the first inequality.

For the second, with $h := |z - z'|$, the symmetric difference of two unit balls at distance h is contained in the closed annulus $\{x : 1 - h \leq |x - z| \leq 1 + h\}$; for $h \leq 1$, $|B_1(z)\Delta B_1(z')| \leq |B_{1+h}| - |B_{(1-h)_+}| \leq n\omega_{n-1}h + O(h^2)$, and for any $h \geq 0$ $|B_1(z)\Delta B_1(z')| \leq 2|B_1| = 2\omega_n$. The clean linear bound $|B_1(z)\Delta B_1(z')| \leq n\omega_{n-1}h$ for all $h \geq 0$ follows from the integral formula $|B_1(z)\Delta B_1(z')| = \int_0^h \mathcal{H}^{n-1}(\partial B_1 \cap \{x : \text{dist}(x, \text{axis}) \leq \dots\}) ds$ bounded above by $\mathcal{H}^{n-1}(\partial B_1) \cdot h = n\omega_{n-1}h$ (the surface area of a unit sphere bounds the rate at which the moving ball sweeps new volume). $\square$

3.3 Localisation of the admissible centres

Lemma 3.3 (Both centres lie in a fixed ball). *Fix a compact $K \subset \mathbb{R}^n$ with $\overline{B_2} \subset K$, say $K = \overline{B_M}$ for some $M \geq 2$. There exists $\eta_1 = \eta_1(n) > 0$ such that for every set E of finite perimeter with $|E| = \omega_n$, $E \subset K$, and $\mathcal{D}(E) \leq \eta_1$, one has $z_F(E) \in K$ and $z_{FJ}(E) \in K$ (where z_F and z_{FJ} are the Fraenkel and FJ minimisers from Theorems 2.1, 2.2).*

Proof. Fraenkel centre. Theorem 2.1 gives $\alpha_{FJ}(E, z_F)^2 \leq C_F \mathcal{D}(E) \leq C_F \eta_1$. Choosing $\eta_1 \leq 1/(4C_F)$ we get $\alpha_{FJ}(E, z_F) \leq \frac{1}{2}$, so $|E \cap B_1(z_F)| \geq |B_1(z_F)| - \frac{1}{2}|B_1(z_F)| = \frac{1}{2}\omega_n$. Since $E \subset K$, this forces $B_1(z_F) \cap K$ to have measure $\geq \frac{1}{2}\omega_n$, which is impossible unless $\overline{B_1(z_F)} \subset \overline{B_{M+1}}$. Hence $|z_F| \leq M + 1$; on enlarging K once and for all to $\overline{B_{M+1}}$ (still satisfying $\overline{B_2} \subset K$) we get $z_F \in K$.

FJ centre. Theorem 2.2 gives $b_{FJ}(E, z_{FJ})^2 \leq C_{FJ} \mathcal{D}(E)$. Apply the change-of-centre identity to α_{FJ} : for any $z' \in K$ with $|z' - z_F| \leq 1$, $\alpha_{FJ}(E, z') \leq \alpha_{FJ}(E, z_F) + n\omega_{n-1}/\omega_n \leq 1$. This gives a uniform Fraenkel bound on a neighbourhood, but does not yet bound $|z_{FJ}|$. Instead, we use that $b_{FJ}(E, z_{FJ})$ is small, so $\partial^* E$ is L^2 -close to a radial vector field centred at z_{FJ} . Theorem 3.4 below (whose proof does not use this lemma) shows $|z_{FJ} - z_F| \leq C\mathcal{D}(E)^{1/2}$. For $\mathcal{D}(E) \leq \eta_1$ small enough, $|z_{FJ}| \leq |z_F| + 1 \leq M + 1$. $\square$

3.4 Centres are close

This is the key quantitative step.

Theorem 3.4 (Fraenkel and FJ centres are close). *There exist $C_{cc} = C_{cc}(n) > 0$ and $\eta_2 = \eta_2(n) > 0$ such that for every set $E \subset \mathbb{R}^n$ of finite perimeter with $|E| = \omega_n$ and $\mathcal{D}(E) \leq \eta_2$,*

$$|z_F(E) - z_{FJ}(E)| \leq C_{cc} \mathcal{D}(E)^{1/2}.$$

Proof. By Theorems 2.1 and 2.2, $\alpha_{FJ}(E, z_F) \leq C_F^{1/2} \mathcal{D}^{1/2}$ and $b_{FJ}(E, z_{FJ}) \leq C_{FJ}^{1/2} \mathcal{D}^{1/2}$. Set $\varepsilon := \mathcal{D}(E)^{1/2}$.

Step 1. Nearly-spherical parametrisation at the FJ centre. By the Fuglede nearly-spherical lemma [4, Lemma 1] as extended to small-deficit sets of finite perimeter in [2, Theorem 3.4], there exists $\eta_0^{\text{Fug}} = \eta_0^{\text{Fug}}(n) > 0$ such that if $\mathcal{D}(E) + b_{FJ}(E, z_{FJ})^2 + \alpha_{FJ}(E, z_{FJ})^2 \leq \eta_0^{\text{Fug}}$, then there exists a Lipschitz function $u : \partial B_1 \rightarrow (-1, \frac{1}{2})$ such that, after translating so that $z_{FJ} = 0$,

$$E = \{(1 + u(\sigma))\sigma : \sigma \in \partial B_1, 0 \leq r \leq 1 + u(\sigma)\},$$

i.e., E is the radial graph of u over the unit sphere centred at z_{FJ} . Moreover $\|u\|_{L^2(\partial B_1)}^2 + \|\nabla_\tau u\|_{L^2}^2 \leq C(n) \mathcal{D}(E)$; the only input is the smallness of the deficit and the smallness of $b_{FJ}(E, z_{FJ})$ and $\alpha_{FJ}(E, z_{FJ})$ (cf. Theorems 2.1, 2.2 and Lemma 3.2; see also [5, Lemma 3.1]).

In particular, for $\mathcal{D}(E)$ small, $\|u\|_{L^2(\partial B_1)} \leq C(n) \varepsilon$ and $\|u\|_\infty \leq C(n) \varepsilon^{1/2}$ by an interpolation between L^2 and H^1 (since for $n \geq 2$, $H^1(\partial B_1) \hookrightarrow L^p(\partial B_1)$ for all $p < \infty$, and L^∞ for $n = 2$ follows from Sobolev embedding; for general $n \geq 2$, $\|u\|_\infty \leq C(n) \varepsilon^{1/2}$ is established in [5, Eq. (2.3)]).

Step 2. First moment of u . Centring shifts of ∂B_1 move the radial graph: for $h \in \mathbb{R}^n$ with $|h|$ small,

$$|E\Delta B_1(h)| = \int_{\partial B_1} |\Phi(u(\sigma), \sigma) - \Phi(\sigma \cdot h, \sigma)| d\mathcal{H}^{n-1}(\sigma) + O(\varepsilon^3),$$

where $\Phi(t, \sigma) := \int_0^{(1+t)} r^{n-1} dr = \frac{(1+t)^n - 1}{n} = t + O(t^2)$, so up to $O(\varepsilon^2)$ error,

$$|E\Delta B_1(h)| = \int_{\partial B_1} |u(\sigma) - \sigma \cdot h| d\mathcal{H}^{n-1}(\sigma) + O(\varepsilon^2).$$

The Fraenkel infimum is attained at $h = z_F$ (after the translation that puts z_{FJ} at the origin). Squaring is more convenient: L^2 -projection of u onto the linear functions $\sigma \mapsto \sigma \cdot h$ on ∂B_1 (which span an n -dimensional subspace, mutually orthogonal in L^2 with $\|\sigma \cdot e_j\|_{L^2}^2 = \omega_n/n$) gives a unique minimiser

$$h^* := \frac{n}{\omega_n} \int_{\partial B_1} \sigma u(\sigma) d\mathcal{H}^{n-1}(\sigma).$$

By Cauchy–Schwarz,

$$|h^*| \leq \frac{n}{\omega_n} (\mathcal{H}^{n-1}(\partial B_1))^{1/2} \|u\|_{L^2(\partial B_1)} = C(n) \|u\|_{L^2} \leq C(n)\varepsilon,$$

i.e., $|h^*| \leq C(n)\mathcal{D}(E)^{1/2}$.

Step 3. The Fraenkel optimum is close to h^ .* The map $h \mapsto |E\Delta B_1(h)|$ is convex up to $O(\varepsilon^2)$ and its quadratic part is the same as that of $h \mapsto \|u - \sigma \cdot h\|_{L^1}$, whose minimiser is h^* up to $O(\varepsilon)$. Quantitatively, for $|h| \leq 1$ small and $\|u\|_{L^2} \leq C\varepsilon$,

$$|E\Delta B_1(h)| - |E\Delta B_1(h^*)| \geq c(n)|h - h^*|^2 - C\varepsilon|h - h^*| \cdot \varepsilon - C\varepsilon^2|h - h^*|,$$

which, evaluated at $h = z_F$ (where the left-hand side is ≤ 0), forces

$$|z_F - h^*| \leq C(n)\varepsilon.$$

Combined with Step 2, $|z_F| \leq |z_F - h^*| + |h^*| \leq C(n)\varepsilon$. Since we translated so $z_{FJ} = 0$, this is exactly $|z_F - z_{FJ}| \leq C_{cc}(n)\mathcal{D}(E)^{1/2}$, as claimed.

Step 4. The smallness threshold η_2 is taken small enough that the hypothesis $\mathcal{D}(E) + b_{FJ}(E, z_{FJ})^2 + \alpha_{FJ}(E, z_{FJ})^2 \leq \eta_0^{\text{Fug}}$ of [2, Thm. 3.4] is satisfied; this uses $\alpha_{FJ}(E, z_{FJ}) \leq \alpha_{FJ}(E, z_F) + (n\omega_{n-1}/\omega_n)|z_{FJ} - z_F|$ (Lemma 3.2) together with the *a posteriori* bound on $|z_{FJ} - z_F|$ just derived; a standard continuity argument shows this hypothesis is self-consistent for sufficiently small $\mathcal{D}(E)$. $\square$

Remark 3.5 (On the constant in Step 3). The convex-quadratic structure of $h \mapsto \|u - \sigma \cdot h\|_{L^2}^2$ gives a clean strictly positive lower bound $c(n) > 0$ at h^* because the linear functions on ∂B_1 form an n -dimensional *closed* subspace of L^2 ; the L^1/L^2 passage uses the fact that, for u small in L^∞ , $|u - \sigma \cdot h|$ and $(u - \sigma \cdot h)^2$ are comparable up to multiplicative $O(\varepsilon^{-1})$ factors only at points where one of them is small; integrated, this is harmless. See [5, Sec. 2–3] and [2, Sec. 3] for the detailed bookkeeping.

3.5 Main theorem

Theorem 3.6 (Joint same-centre Fusco–Julin stability). *For every $n \geq 2$ and every compact set $K \subset \mathbb{R}^n$ with $\overline{B_2} \subset K$, there exist $C_{\text{scFJ}} = C_{\text{scFJ}}(n, K) > 0$ and $\eta_0 = \eta_0(n, K) > 0$ such that for every set $E \subset \mathbb{R}^n$ of finite perimeter with $|E| = \omega_n$, $E \subset K$, and $\mathcal{D}(E) \leq \eta_0$, the minimum*

$$m_E := \min_{z \in K} \Phi_E(z)$$

is attained, and

$$m_E \leq C_{\text{scFJ}} \mathcal{D}(E).$$

In particular there exists $z_E \in K$ with

$$\alpha_{\text{FJ}}(E, z_E)^2 + b_{\text{FJ}}(E, z_E)^2 \leq C_{\text{scFJ}} \mathcal{D}(E).$$

Proof. Attainment. By Lemma 3.1, Φ_E is continuous on K , and K is compact; hence the minimum is attained.

Upper bound at z_{FJ} . Let z_F, z_{FJ} be the Fraenkel and FJ centres of Theorems 2.1, 2.2. By Lemma 3.3 (after choosing η_0 small), both lie in K . By Theorem 2.2, $b_{\text{FJ}}(E, z_{\text{FJ}})^2 \leq C_{\text{FJ}} \mathcal{D}(E)$. By Lemma 3.2 and Theorem 3.4,

$$\alpha_{\text{FJ}}(E, z_{\text{FJ}}) \leq \alpha_{\text{FJ}}(E, z_F) + \frac{n\omega_{n-1}}{\omega_n} |z_{\text{FJ}} - z_F| \leq C_F^{1/2} \mathcal{D}(E)^{1/2} + \frac{n\omega_{n-1}}{\omega_n} C_{cc} \mathcal{D}(E)^{1/2}.$$

Squaring,

$$\alpha_{\text{FJ}}(E, z_{\text{FJ}})^2 \leq C'(n) \mathcal{D}(E).$$

Summing,

$$\Phi_E(z_{\text{FJ}}) = \alpha_{\text{FJ}}(E, z_{\text{FJ}})^2 + b_{\text{FJ}}(E, z_{\text{FJ}})^2 \leq (C'(n) + C_{\text{FJ}}) \mathcal{D}(E) =: C_{\text{scFJ}}(n) \mathcal{D}(E).$$

Since $z_{\text{FJ}} \in K$, $m_E = \min_K \Phi_E \leq \Phi_E(z_{\text{FJ}}) \leq C_{\text{scFJ}} \mathcal{D}(E)$, as claimed. $\square$

Remark 3.7 (On the constants). The constant $C_{\text{scFJ}}(n, K)$ does not depend on K in the proof above; the dependence on K is only through the containment condition $E \subset K$ (used in Lemma 3.3), which we encode by allowing η_0 to depend on K so that the Fraenkel and FJ centres lie in K . Thus we may write $C_{\text{scFJ}} = C_{\text{scFJ}}(n)$ once K is fixed.

4 Borel measurable selection on torsion levels

We now assemble the measurable-selection input. The setting is $E_\rho := \{u > t(\rho)\} \subset B_R$, with $|E_\rho| = \omega_n \rho^n$, $\rho \in [\rho_*, 1]$, and isoperimetric defect $\mathcal{I}(\rho) := D_I(t(\rho)) = (P(E_\rho) - P(B_\rho))(P(E_\rho) + P(B_\rho))$. At good (a.e. finite-perimeter) levels $P(E_\rho) - P(B_\rho) \leq C(n, \rho_*) \mathcal{I}(\rho)$.

For the joint functional we use the *scale-normalised* Φ : writing $\hat{E}_\rho := \rho^{-1} E_\rho$ (unit volume), define

$$\Phi(\rho, z) := \Phi_{\hat{E}_\rho}(z) = \alpha_{\text{FJ}}(\hat{E}_\rho, z)^2 + b_{\text{FJ}}(\hat{E}_\rho, z)^2.$$

Equivalently, for $z \in K$, $\Phi(\rho, \rho z) = \alpha_{\text{FJ}}(E_\rho, \rho z)^2 + b_{\text{FJ}}(E_\rho, \rho z)^2$ by the scaling Lemma in `WIP_FJCenterBridge.tex`.

Lemma 4.1 (Joint Borel/continuity of Φ). *Assume the family $\rho \mapsto E_\rho$ is given via the torsion coarea: $\rho \mapsto \mathbf{1}_{E_\rho}$ is Borel from $[\rho_*, 1]$ into $L^1(\mathbb{R}^n)$, and $\rho \mapsto \partial^* E_\rho$ admits a Borel parametrisation in the sense that the pushforward measures $(\rho \mapsto \mathcal{H}^{n-1} \llcorner \partial^* E_\rho)$ form a Borel field of finite Radon measures on $\mathbb{R}^n$ (cf. [7, §15]). Then $(\rho, z) \mapsto \Phi(\rho, z)$ is Borel measurable on $\{\rho : P(E_\rho) < \infty\} \times K$, and continuous in z for each fixed ρ .*

Proof. Borel in ρ , continuous in z . By Lemma 3.1, $z \mapsto \Phi(\rho, z)$ is continuous on K for each ρ with $P(E_\rho) < \infty$. For Borel measurability in ρ , note that:

1. $\rho \mapsto \alpha_{\text{FJ}}(E_\rho, \rho z)^2 = (\|\mathbf{1}_{E_\rho} - \mathbf{1}_{B_\rho(\rho z)}\|_{L^1} / |B_\rho|)^2$ is Borel: L^1 -norms of Borel L^1 -valued maps are Borel, and the translation $\rho \mapsto \mathbf{1}_{B_\rho(\rho z)}$ is continuous.
2. $\rho \mapsto b_{\text{FJ}}(E_\rho, \rho z)^2 = \frac{1}{2} \int |\nu_{E_\rho} - (x - \rho z) / |x - \rho z||^2 d\mathcal{H}^{n-1} \llcorner \partial^* E_\rho$ is Borel: it is the integral of a bounded continuous function on $\mathbb{R}^n \setminus \{\rho z\}$ against a Borel field of finite Radon measures [7, Prop. 15.10].

Sums, ratios, and squares of Borel functions are Borel, and joint Borel measurability $(\rho, z) \mapsto \Phi(\rho, z)$ follows from Borel-in- ρ + continuous-in- z (see e.g. [1, Lem. III.14]; alternatively sample on a countable dense net in K). $\square$

Proposition 4.2 (Borel same-centre selection). *Let $R \geq 1$, $\rho_* \in (0, 1)$, and $K := \overline{B_{R+1}} \subset \mathbb{R}^n$ (so $\overline{B_2} \subset K$). Set*

$$G_I := \{\rho \in [\rho_*, 1] : \mathcal{I}(\rho) \leq \eta_I\},$$

where $\eta_I = \eta_I(n, R, \rho_*) > 0$ is small enough that Theorem 3.6 applies to $\hat{E}_\rho = \rho^{-1}E_\rho$ on G_I . Then there exist $C = C(n, R, \rho_*) > 0$ and a Borel map $\rho \mapsto z_\rho \in K$ defined on $[\rho_*, 1]$ (with the convention $z_\rho := 0$ for $\rho \notin G_I$) such that for every $\rho \in G_I$,

$$\alpha_{\text{FJ}}(\hat{E}_\rho, z_\rho/\rho)^2 + b_{\text{FJ}}(\hat{E}_\rho, z_\rho/\rho)^2 \leq C_{\text{scFJ}} \mathcal{D}(\hat{E}_\rho) \leq C \mathcal{I}(\rho),$$

and in particular

$$|E_\rho \Delta B_\rho(z_\rho)|^2 + \int_{\partial^* E_\rho} \left| \nu_{E_\rho} - \frac{x - z_\rho}{|x - z_\rho|} \right|^2 d\mathcal{H}^{n-1} \leq C \mathcal{I}(\rho).$$

Proof. Fix $C_0 := 2C_{\text{scFJ}}(n)$, and define the multifunction

$$\Gamma : G_I \rightrightarrows K, \quad \Gamma(\rho) := \{z \in K : \Phi(\rho, z/\rho) \leq C_0 \mathcal{D}(\hat{E}_\rho)\}.$$

By Theorem 3.6, $\Gamma(\rho) \neq \emptyset$ for every $\rho \in G_I$. By Lemma 3.1 (continuity in the centre) and the continuity of the map $z \mapsto z/\rho$, $\Gamma(\rho)$ is closed in K , hence compact.

Borel measurability of the graph. The graph $\text{Gr}(\Gamma) = \{(\rho, z) \in G_I \times K : \Phi(\rho, z/\rho) \leq C_0 \mathcal{D}(\hat{E}_\rho)\}$ is Borel by Lemma 4.1: the inequality involves Borel functions of (ρ, z) (and $\rho \mapsto \mathcal{D}(\hat{E}_\rho)$ is Borel, being a Borel function of $P(E_\rho)$).

Kuratowski–Ryll–Nardzewski selector. The closed-valued multifunction Γ on the Borel space G_I with values in the Polish space K has Borel graph, hence is Borel measurable in the sense of Kuratowski–Ryll–Nardzewski (cf. [1, Theorem III.6 and Theorem III.30]); by the same theorem, there exists a Borel selector $\rho \mapsto z_\rho \in \Gamma(\rho)$. Extending by $z_\rho := 0$ on $[\rho_*, 1] \setminus G_I$ preserves Borel measurability.

Quantitative estimates. For $\rho \in G_I$, $\mathcal{D}(\hat{E}_\rho) = \mathcal{D}(E_\rho) = (P(E_\rho) - P(B_\rho))/P(B_\rho) \leq C(n, \rho_*) \mathcal{I}(\rho)$. The joint inequality $\Phi_{\hat{E}_\rho}(z_\rho/\rho) \leq C_0 \mathcal{D}(\hat{E}_\rho) \leq C \mathcal{I}(\rho)$ holds by definition of Γ . Multiplying out scales (Lemma in `WIP_FJCenterBridge.tex`, restated as $|E_\rho \Delta B_\rho(z_\rho)| = \rho^n |\hat{E}_\rho \Delta B_1(z_\rho/\rho)|$ and similarly for the normal-oscillation integral, using $P(E_\rho)/P(B_\rho) \leq 1 + C \mathcal{D}(E_\rho)$), one obtains the two unnormalised displayed bounds. $\square$

5 Corollary for the torsion application

Corollary 5.1 (Replacement for the conditional centre bridge). *Let $\Omega \subset B_R$ be the bounded torsion domain of the Plan 2 setting, $E_\rho = \{u > t(\rho)\}$ its volume-radius parametrised super-levels, $\rho_* \in (0, 1)$, and $K := \overline{B_{R+1}}$. For every ρ in the good set G_I of Proposition 4.2 (with $\eta_I = \eta_I(n, R, \rho_*)$ chosen so that Theorem 3.6 applies), the centre $z_\rho \in K$ selected by Proposition 4.2 is Borel in ρ and satisfies the three conclusions of the good-level centre proposition of `WIP_FJCenterBridge.tex` (Proposition 3.1 there, “Good-level same-centre estimates”), with the same constant $C = C(n, R, \rho_*)$. In particular,*

1. $|z_\rho| \leq R + 1$,
2. $|E_\rho \Delta B_\rho(z_\rho)|^2 \leq C \mathcal{I}(\rho)$,
3. $\int_{\partial^* E_\rho} |\nu_\rho - (x - z_\rho)/|x - z_\rho||^2 d\mathcal{H}^{n-1} \leq C \mathcal{I}(\rho)$.

Proof. The first conclusion is by construction ($z_\rho \in K$). The remaining two are the unnormalised forms of the joint estimate in Proposition 4.2, obtained by writing $\alpha_{\text{FJ}}(\hat{E}_\rho, z_\rho/\rho)^2 = (|E_\rho \Delta B_\rho(z_\rho)|/|B_\rho|)^2$, multiplying by $|B_\rho|^2$, and using $\rho \geq \rho_*$; and by writing $b_{\text{FJ}}(\hat{E}_\rho, z_\rho/\rho)^2 \cdot P_{\hat{E}_\rho}^* = \frac{1}{2}\rho^{-(n-1)} \int_{\partial^* E_\rho} |\nu - e|^2 d\mathcal{H}^{n-1}$, multiplying out, and using $P(B_\rho) = n\omega_n \rho^{n-1}$, $P(E_\rho) \leq 2P(B_\rho)$ on G_I . $\square$

Remark 5.2 (What is unconditional, and what is cited). **Unconditional.** Theorems 3.6, 3.4, Lemma 3.1, Lemma 3.2, Lemma 3.3, Lemma 4.1, and Proposition 4.2. Each is proved here from the following *cited* inputs:

- Theorem 2.1 (sharp Fraenkel asymmetry stability): [3, 6].
- Theorem 2.2 (sharp normal-oscillation stability): [5].
- Fuglede’s nearly-spherical lemma [4] and its extension to small-deficit finite perimeter sets [2, Theorem 3.4], used in Theorem 3.4 (Step 1).
- Coarea Borel structure of reduced boundaries [7, §15], used in Lemma 4.1.
- Kuratowski–Ryll–Nardzewski measurable selection [1, Theorem III.6, III.30], used in Proposition 4.2.

Remaining external dependency. The convex-quadratic inequality of Theorem 3.4, Step 3 (Remark 3.5), uses two standard ingredients of the Fuglede/Fusco–Julin analysis (an L^2 -orthogonality of the sphere harmonics of degree 1 and an L^∞ -bound on the nearly-spherical parametrisation); these are spelled out in [5, Sec. 2–3] and [2, Sec. 3] but are not reproduced here. This is the only step of the present file that is not internally self-contained.

References

- [1] C. Castaing, M. Valadier, *Convex Analysis and Measurable Multifunctions*, Lecture Notes in Math. 580, Springer, 1977.
- [2] M. Cicalese, G.P. Leonardi, A selection principle for the sharp quantitative isoperimetric inequality, *Arch. Ration. Mech. Anal.* **206** (2012), 617–643.
- [3] A. Figalli, F. Maggi, A. Pratelli, A mass transportation approach to quantitative isoperimetric inequalities, *Invent. Math.* **182** (2010), 167–211.
- [4] B. Fuglede, Stability in the isoperimetric problem for convex or nearly spherical domains in $\mathbb{R}^n$, *Trans. Amer. Math. Soc.* **314** (1989), 619–638.
- [5] N. Fusco, V. Julin, A strong form of the quantitative isoperimetric inequality, *Calc. Var. Partial Differential Equations* **50** (2014), 925–937.
- [6] N. Fusco, F. Maggi, A. Pratelli, The sharp quantitative isoperimetric inequality, *Ann. of Math.* **168** (2008), 941–980.
- [7] F. Maggi, *Sets of Finite Perimeter and Geometric Variational Problems*, Cambridge Stud. Adv. Math. 135, Cambridge Univ. Press, 2012.